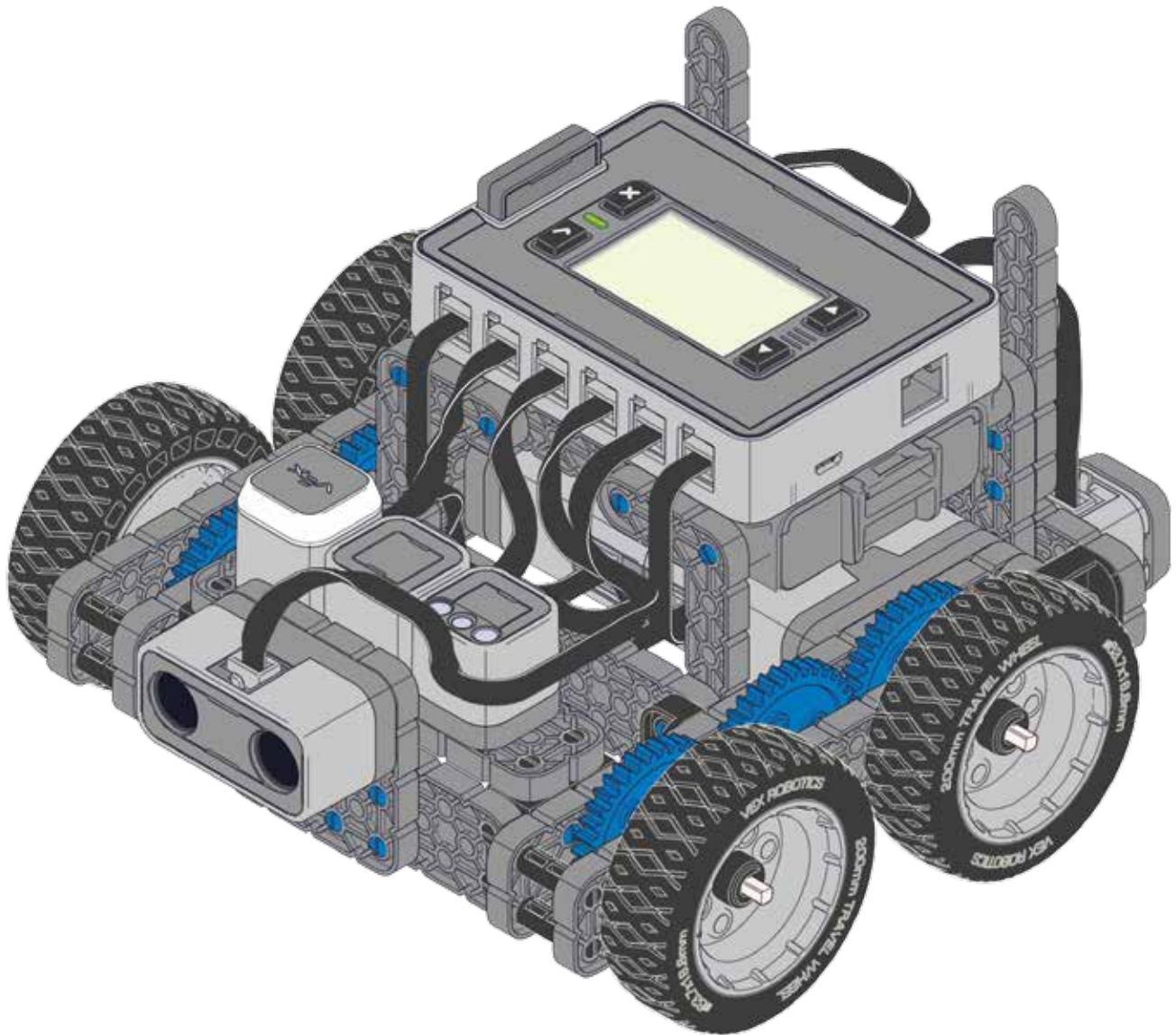
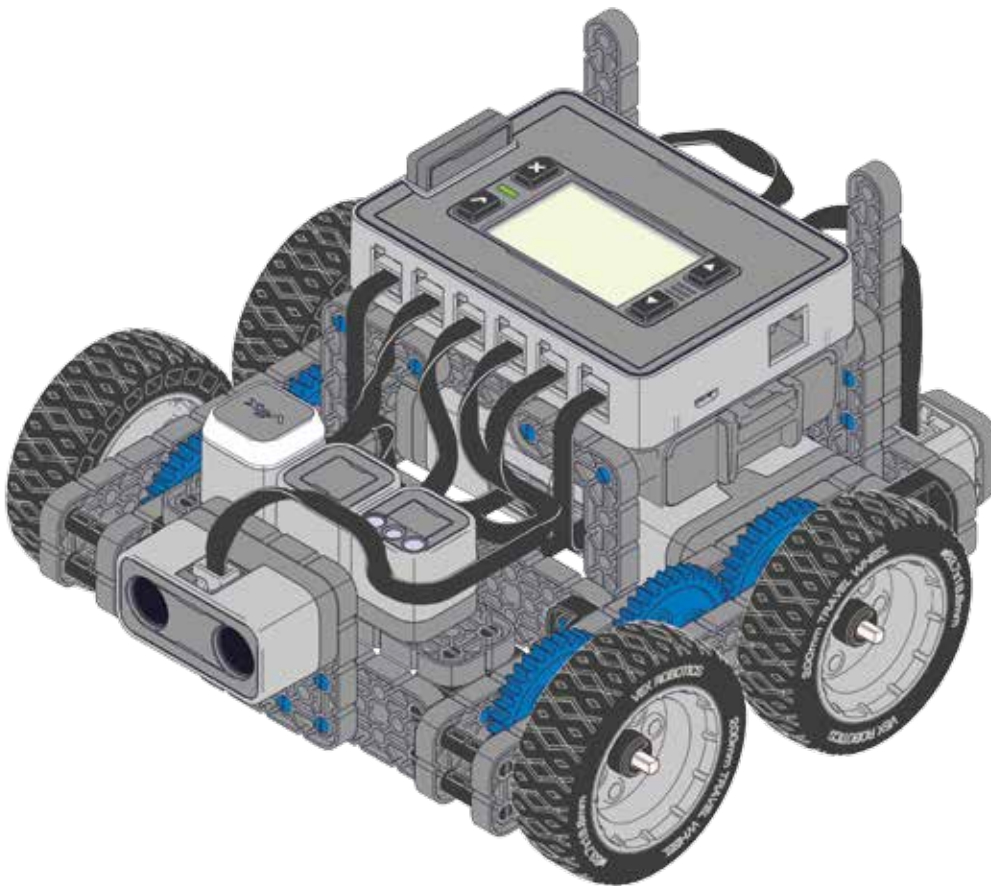




Autopilot Robot





I.cl1

Overview of Activity:

Campers will build the VEX IQ Autopilot robot and run it with particular sensors connected or disconnected to gain an understanding of how sensors affect the “intelligence” of a robot. They will conduct mini investigations into robot behavior using the Autopilot test modes.

Learning Objectives:

- Campers will learn key terminology related to sensors and programming.
- Campers will gain an understanding of how sensors and motors contribute to the capabilities of a robot.
- Campers will learn sensor types and functions.
- Campers will begin to think critically and creatively about how sensors can be incorporated into a range of robotics tasks.

Suggested Timing:

Up to 2 hours

- This activity may take less time if campers have previous experience with sensors and robotics.
- This activity may take more time if campers have not already built the Standard Drive Base from Learning Activity F – Standard Drive Base.

I.cl1 cont.

VEX IQ Materials:

- VEX IQ Super Kit
- Completed Standard Drive Base
- Highrise Challenge Field (half or full size) and Cubes
- Empty VEX IQ Super Kit box
- VEX IQ Build Instructions - Autopilot Instructions (Kit Documentation)  **1+4**
- VEX IQ Control System User Guide – 6.2 Autopilot Program (Kit Documentation)  **6.2**
-  Autopilot Investigation Handout
-  VEX IQ Sensor Overview

Optional Materials:

- Shoes, text books, plastic cups and other objects of different weights to act as barriers or obstacles

Pre-learning Suggestions:

- If possible, complete Learning Activity F – Standard Drive Base.

Based on VEX IQ Robotics Education Guide:

- Running Autopilot Modes (I.6)

I.cl2

Detailed Directions

1. Provide campers with the kit documentation, VEX IQ Build Instructions - Autopilot Instructions and give them time to complete the construction of their robots.
2. When the robot has been assembled, challenge campers to identify each sensor that has been added and brainstorm a few tasks that sensor could help to achieve. Encourage campers to note their ideas in a notebook for future use.

If needed, provide campers with "VEX IQ Sensor Overview" Background Information for more specific details about sensor names, specs, use, and default functionality (how it works without any additional programming).
3. Tell campers that they will be experimenting with three different autopilot programs. Have them set up a half-size Highrise Challenge Field or block off a 2'x 4' area using tables, boxes or other barriers to contain the robot. This will be the testing area for a few different investigations.
4. Provide campers with the kit documentation, VEX IQ Control System User Guide – 6.2 Autopilot Program. Tell them that they will need to use this robot and program to complete their investigation.
5. Provide campers with the handout, Autopilot Investigation. They should complete their investigations with the robots on the field (or other designated space) and conduct several tests for each investigation.

6. Conduct the investigations.

INVESTIGATION #1: Explore the Autopilot test modes.

See I.cs1 for steps.

Question to ask campers:

- Which sensors are being used to create each behavior?

Descriptions of robot behavior when running the Autopilot Program can be found in the kit documentation, VEX IQ Control System User Guide, 6.2 Autopilot Program.

INVESTIGATION#2: Unplug the VEX IQ Distance Sensor.

See I.cs1 for steps.

Questions to ask campers:

- What tasks is the robot able to complete without the Distance Sensor?
- Is it possible to stop the robot without the Distance Sensor? (yes)
- What stops the robot when the Distance Sensor is disconnected? (Force)
- How does the robot know it needs to back up? Is there a sensor being used or something else? (The smart motors provide feedback based on force.)
- How much force is required to provide the feedback to the robot to stop and change directions? (What is the threshold?)



Note: This is a great point to introduce the concept of threshold, an important point when programming sensors, to campers at a high level.

Exact measurements of force are not required, but campers should consider the weight or amount of friction of an object and how this affects the feedback a robot receives from its sensors or motor. The concept of threshold is used to set ranges for sensor feedback that determine how a robot will behave in response to stimuli. (e.g., If X amount or less of force is detected, the robot will keep moving. If X amount of force or more is detected, the robot will back up.)

INVESTIGATION #3: Unplug the other VEX IQ Sensors.

See I.cs1 for steps.

Questions to ask campers:

- Will the robot function when each sensor is unplugged?
- Which sensors affect the programming the most?
- How do you think each sensor functions in the Autopilot Program?
- For each sensor function, could a different sensor be used to perform a similar function?

7. If time allows, place a variety of objects (e.g., stacks of Highrise cubes, structures and archways made from VEX components) on the competition field and run the Autopilot program with all sensors for one minute.

Questions to ask campers:

- How many obstacles/structures remain?
- Were any of the modes more effective at avoiding the obstacles?
- Which obstacles were easiest to avoid? Why?

Tips and Best Practices:

- When selecting objects to place on the Challenge Field, keep in mind that the Color Sensor will automatically detect and respond to certain colors when they are near the Color Sensor. (e.g., red, green, blue).

Extension:

- Move the Color Sensor to a forward-facing position on the robot. Experiment with programming to replicate the movement patterns completed by the Autopilot Program and investigate different light and color thresholds to make contact with objects that are outside a given range and avoid objects within it.

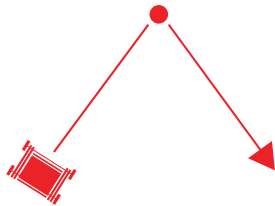


Autopilot Investigation

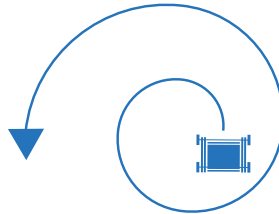
Student Name(s): _____

Instructions:

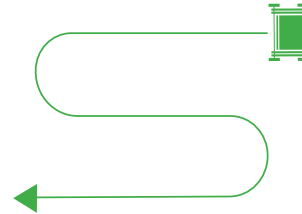
Build the Autopilot Robot and run the Autopilot Program to conduct the investigations below.



Random Mode



Spiral Mode



Lawnmower Mode

INVESTIGATION #1: Explore the Autopilot test modes.

1. Place the Autopilot robot on the Challenge Field and start the program.
2. Place your hand in front of the robot. Describe how the robot reacts.
3. Place different colored Highrise cubes in front of the robot. Describe how the robot reacts to each color.

Draw or describe your observations here

INVESTIGATION#2: Unplug the VEX IQ Distance Sensor.

1. Disconnect the VEX IQ Distance Sensor.
2. Make a prediction about how removing this sensor will affect the robot's ability to stop or change directions.
3. Place the Autopilot robot on the Challenge Field and start the program. Describe how the robot's behavior is the same or different from INVESTIGATION 1.
4. Place different objects in front of the robot as it completes the Autopilot Program.
 - A Highrise Cube. Describe what happens.
 - A running shoe or something of similar weight. Describe what happens.
 - An empty VEX IQ Super Kit box.
5. Place the robot near a wall and run the Autopilot program. Describe what happens.

Draw or describe your observations here

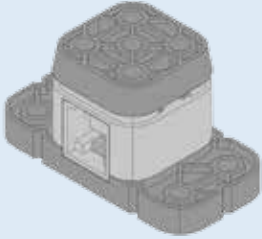
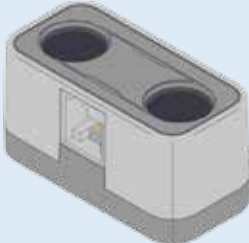
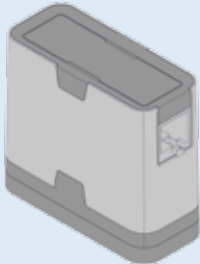
INVESTIGATION #3: Unplug the other VEX IQ Sensors.

1. Reconnect the VEX IQ Distance Sensor.
2. Unplug each VEX IQ Sensor one after another and test what happens to the Autopilot robot's behavior.
3. Describe what happens when each sensor is unplugged.

Draw or describe your observations here



VEX IQ Sensor Overview

Sensor Name	Sensor Image	Sensor Specs & Use	Default Functionality
Bumper Switch		Allows for the sense of touch in the VEX IQ platform. Detect a wall, obstacle, or limit mechanism movement.	With a Bumper Switch in port 2, it disables/enables a Smart Motor in port 4.
Touch LED		Smart Sensor with red, green, blue LEDs. Constant on, off, or blink at any desired rate. Touch sensor with finger for interaction.	Enables and disables Autopilot or similar robot running Driver Control Program when dome of sensor is tapped. Glows green when enabled, red when disabled.
Distance Sensor		Uses ultrasonic sound waves to measure distance. Measures distance from 1 inch to 10 feet. Commonly used to avoid obstacles.	Slows down and eventually stops as Autopilot or similar robot running Driver Control Program as the robot approaches an obstacle.
Color Sensor		Detects the color of objects. Measures independent red, green, and blue in 256 levels each.	Enables and disables Autopilot or similar robot running Driver Control Program when the sensor is "shown" a green card (enabled) or red card (disabled).
Gyro Sensor		Measures turn rate and calculates direction. Frequently used in autonomous robot navigation and turning.	With Autopilot or similar robot running Driver Control Program, it returns a robot to its original direction when driving stops.
Smart Motor		Commands and measures speed, direction, time, revolutions and/or degrees of turn using its Encoder.	No extra default functionality, but Encoders allow for superior motor control with simple programming.